PROJECT TITLE:

HOSPITAL EMERGENCY ROOM ANALYSIS

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TOOLS: SQL, EXCEL

DATE : 19-04-2026

Business Problem

The hospital emergency room is facing challenges in managing patient flow, reducing wait times, and maintaining patient satisfaction.

There is limited visibility into:

• Daily patient volume

• Waiting time trends

• Patient satisfaction levels

This affects overall efficiency and service quality.

Objective

To build an Emergency Room Analysis Dashboard to:

• Monitor patient volume
• Reduce waiting time
• Improve patient satisfaction
• Support better decision-making

Scope of Analysis

This project analyses Emergency Room data to evaluate hospital performance and patient service efficiency.

Analysis of patient volume trends
• Evaluation of average wait time
• Assessment of patient satisfaction levels
• Identification of peak hours and busy periods
• Overall performance analysis of emergency room services

Current State (As-Is Process)

The current emergency room process lacks efficiency due to manual handling and a lack of real-time monitoring.

Patients arrive → Manual registration → Delays in processing → Long waiting times → Low patient satisfaction

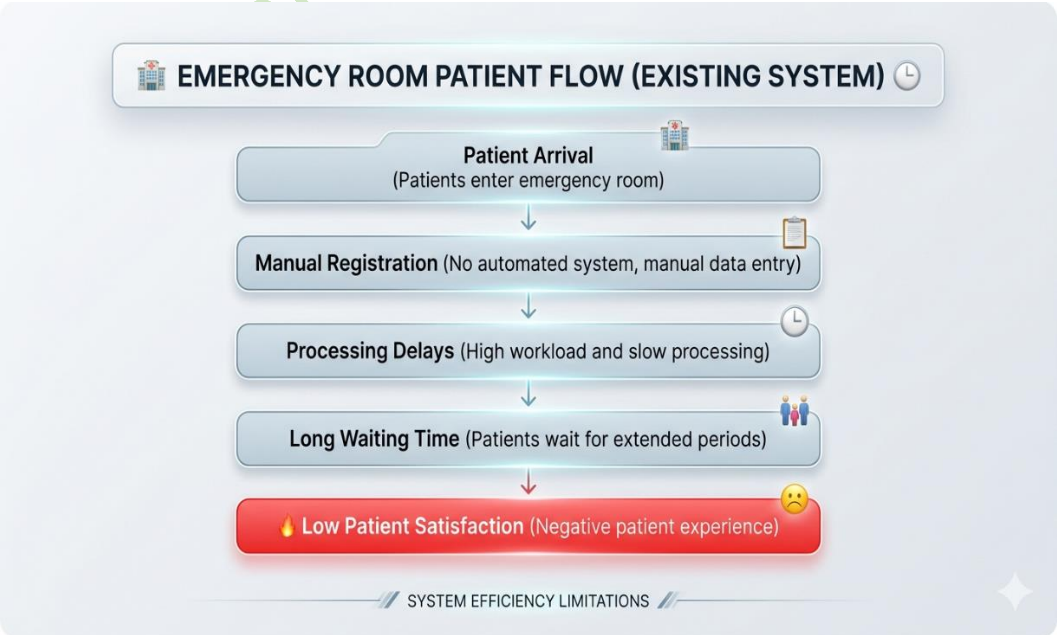
Requirement Gathering

Stakeholders:

• Hospital Management
• Doctors & Medical Staff
• Operations Team

JAD Sessions were conducted to identify key challenges:

• High patient load during peak hours
• Long waiting times affecting service quality
• Lack of a real-time monitoring and tracking system



Business Requirements (BRD)

The hospital aims to improve emergency room performance by focusing on operational efficiency, patient experience, and data-driven monitoring.

• Improve ER efficiency – Optimise emergency room operations to handle patient flow effectively and reduce bottlenecks

• Reduce patient wait time – Minimise delays between patient arrival and treatment to improve service speed

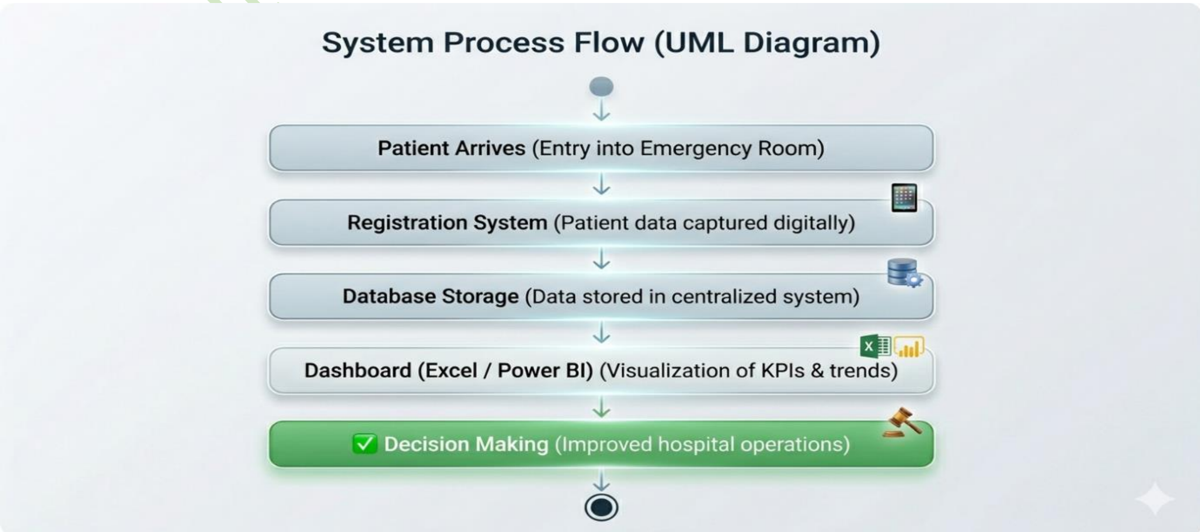
• Enhance patient experience – Improve overall service quality and patient satisfaction levels

• Monitor hospital performance – Track key performance indicators for better decision-making

Future State (To-Be Process)

The improved system focuses on automation, real-time tracking, and better decision-making support.

Patient arrives → Data is recorded digitally → Stored in centralized system → Real-time dashboard updates → Faster and informed decision-making



Functional Requirements (FRD)

The dashboard provides real-time insights into emergency room performance to support effective decision-making.

The system should:

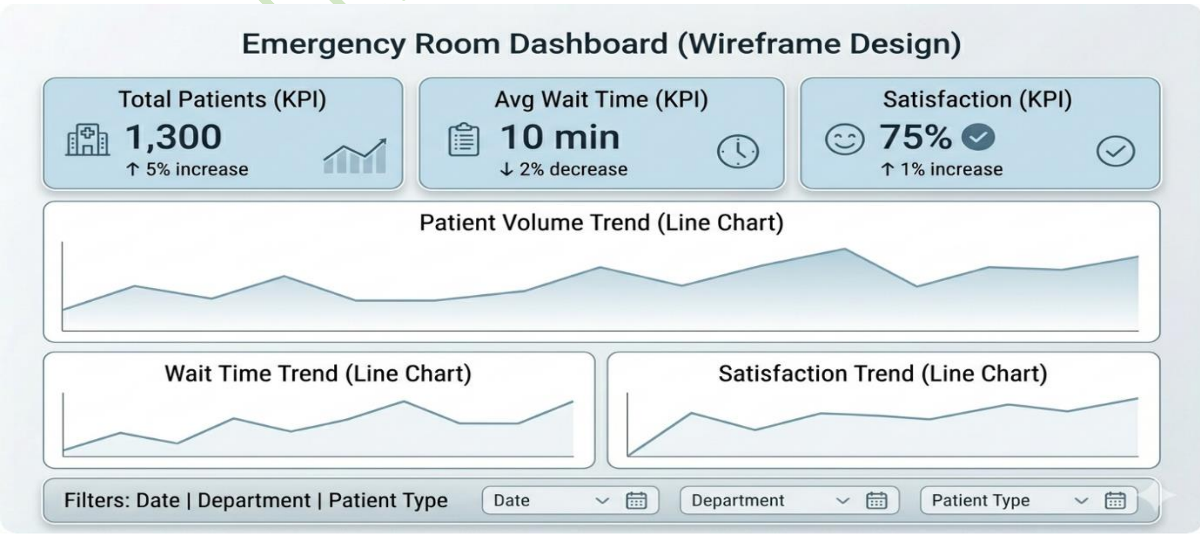
- Display total number of patients
- Show average wait time
- Track patient satisfaction score
- Provide daily trend analysis
- Include interactive filters (date, department)

Key Performance Indicators (KPIs)

• Number of Patients – Represents total patient visits, helping identify peak hours and workload patterns

• Average Wait Time – Indicates operational efficiency and helps detect delays in service

• Patient Satisfaction Score – Reflects overall patient experience and service quality



****The wireframe was designed to ensure clear visibility of key KPIs, trends, and filters, enabling quick and effective decision-making****

Data Overview & Analysis Framework

Dataset Description

The dataset includes information about emergency room patient visits, focusing on visit details, waiting time, and satisfaction scores to support performance analysis.

Data Dictionary

- Patient_ID – Unique identifier assigned to each patient
- Visit_Date – Date of emergency room visit
- Wait_Time – Time taken before receiving treatment
- Satisfaction_Score – Patient feedback rating indicating service quality

System Process Flow

Patient → System → Database → SQL Analysis → Dashboard → Insights → Decision Making

KPI Calculation

- Total Patients = COUNT(Patient_ID)
- Average Wait Time = AVG(Wait_Time)
- Satisfaction Score = AVG(Satisfaction_Score)

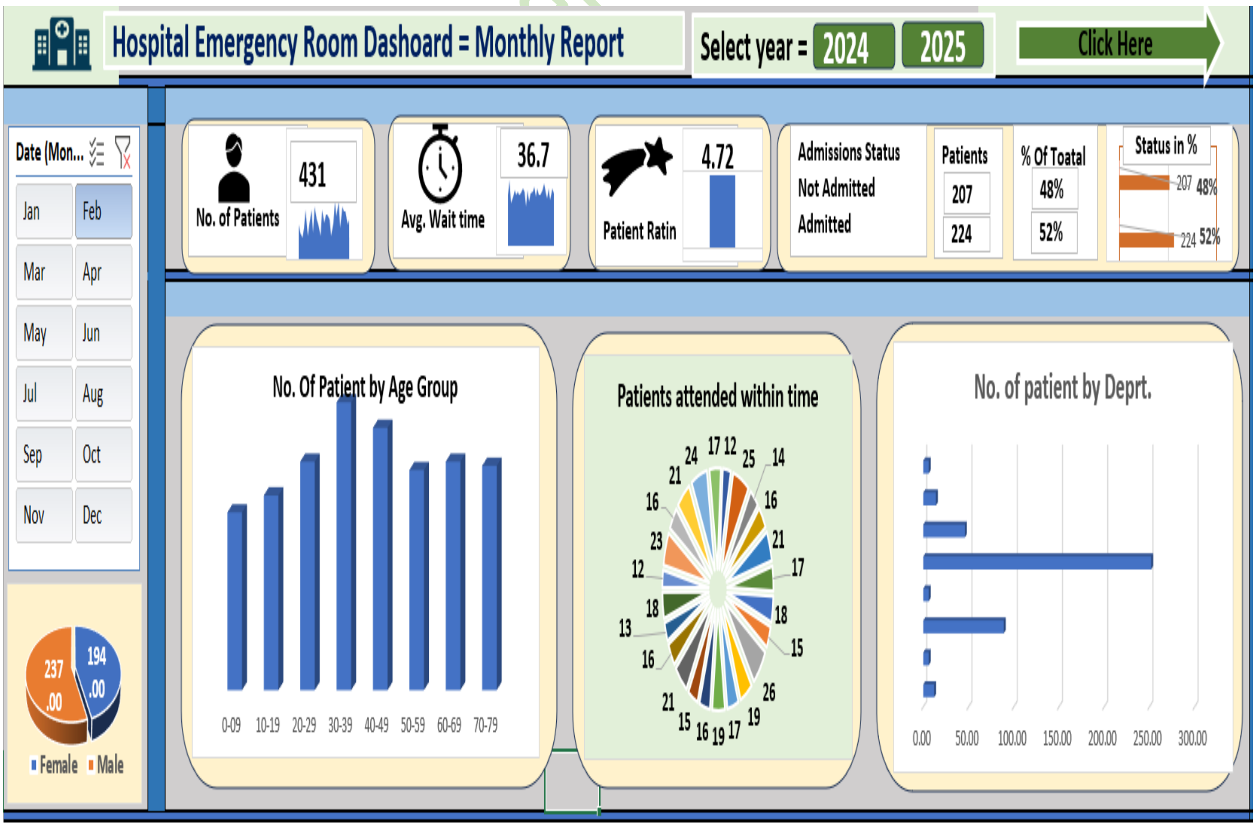
SQL Analysis

- Analyse total patients per day to identify daily trends
- Calculate average wait time to measure efficiency
- Evaluate patient satisfaction scores for service quality
- Identify peak days and high patient load periods

Dashboard Overview

- KPI Cards – Patients, Wait Time, Satisfaction
- Trend Charts – Patient volume and performance trends
- Filters – Date and other parameters for detailed analysis

Emergency Room Dashboard



Insights

- Peak days experience significantly higher patient volume
- Increased patient load leads to longer waiting times
- Patient satisfaction tends to decline during busy periods
- Variations in daily trends highlight operational inefficiencies

Recommendations

- Increase staff availability during peak hours to manage workload
- Optimize patient flow to reduce delays and improve efficiency
- Implement better scheduling strategies to minimize wait time
- Focus on improving service quality during high-demand periods

Conclusion

This project demonstrates how data analysis and visualization can enhance emergency room efficiency, improve patient experience, and support informed decision-making through real-time insights.

Tools & Assumptions

Tools Used

- Microsoft Excel – Used for data cleaning, analysis, and dashboard development
- SQL – Used for data extraction, transformation, and analysis
- Data Visualisation – Used to present insights in a clear and interactive manner

Assumptions

- The dataset used is accurate and complete
- Wait time is properly recorded without errors

Thank You…….